5	**(a)**	**(i)**	sum of currents into a junction = sum of currents out of junction	B1	[1]
		(ii)	charge	B1	[1]
	(b)	**(i)**	$\Sigma E = \Sigma IR$		
			$20 - 12 = 2.0(0.6 + R)$ (*not used 3 resistors 0/2*)	C1	
			$R = 3.4\,\Omega$	A1	[2]
		(ii)	$P = EI$	C1	
			$= 20 \times 2$		
			$= 40\,\text{W}$	A1	[2]
		(iii)	$P = I^2R$	C1	
			$P = (2)^2 \times (0.1 + 0.5 + 3.4)$		
			$= 16\,\text{W}$	A1	[2]
		(iv)	efficiency = useful power / output power	C1	
			24 / 40 = 0.6 or 12 × 2 / 20 × 2 or 60%	A1	[2]

6 (a) (i) diffraction bending/spreading of light at edge/slit B1